

HILTON REVENUE MANAGEMENT & PRICING OPTIMIZATION ANALYSIS

Data Analytics Project Report

Project Type: Hypothetical Business Case Study
Industry: Hospitality & Revenue Management
Tools Used: Python, MySQL, Power BI
Dataset Size: 50,000+ Records | 5 Analytical Tables

1. Introduction

The hospitality industry depends on effective revenue management strategies to optimize room pricing, maximize occupancy, and improve customer experience. Hotels must analyze booking patterns, seasonal demand, market pricing, and guest feedback to make informed pricing decisions.

This project focuses on analyzing Hilton hotel performance data to identify key revenue drivers, pricing opportunities, demand patterns, and customer experience factors.

The objective of this analysis was to:

- Identify high-performing room categories and revenue sources
- Analyze seasonal demand and pricing opportunities
- Evaluate Hilton’s competitive pricing position
- Understand event-driven demand changes
- Determine the impact of guest satisfaction on booking value

The project follows an end-to-end analytics approach using Python, SQL, and Power BI.

2. Dataset Overview

A synthetic hospitality dataset was created to simulate real-world hotel operations.

Attribute	Value
Records	50,000+
Tables	5
Data Type	Structured Data

The dataset included:

Table	Purpose
Bookings	Revenue, ADR, room type, booking channel, guest segment
Hotel KPI Daily	Occupancy, seasonality, demand indicators
Competitor Pricing	Market ADR comparison

Guest Reviews	Ratings and customer sentiment
Market Events	Events influencing demand

3. Data Cleaning & Preparation

Data preprocessing was performed using Python and Pandas.

Key activities included:

- Data quality checks
- Date format standardization
- Missing value validation
- Duplicate record checks
- Categorical data standardization
- Feature creation for booking month and season analysis

The cleaned datasets were imported into MySQL using SQLAlchemy and PyMySQL for further business analysis.

4. SQL Business Analysis

SQL was used to answer key revenue management questions and generate actionable business insights.

Analysis 1: Which room types generate the highest revenue and contribute most to profitability?

Objective: Identify the room categories contributing the highest revenue and profitability.

Insight: Ocean View Rooms generated the highest revenue (**\$22.87M; 28.92% of total revenue**) from **15,022 bookings**, while Suites achieved the highest ADR (**\$734.11**) despite only **5,029 bookings**, highlighting premium pricing opportunities.

	Room_Type	Total_Bookings	Total_Revenue	Average_Room_Rate	Revenue_Contribution_Percentage
▶	Ocean View Room	15022	22873907	384.72	28.92
	Standard Room	19855	22390808	284.97	28.31
	Ocean Front Room	10094	19197690	484.90	24.28
	Suite	5029	14619163	734.11	18.49

Analysis 2: How does hotel revenue performance change by month and season?

Objective: Analyze seasonal revenue, occupancy, and ADR trends.

Insight: Peak Season consistently outperformed all other periods, with occupancy above **88%**, ADR reaching **\$488.75**, and monthly revenue exceeding **\$77.03M**, while Low Season revenue declined to **\$24.03M–\$26.54M**.

	Revenue_Month	Season	Average_Occupancy_Percentage	Average_ADR	Total_Revenue	Average_RevPAR
►	August	Peak Season	89.71	480.13	77025850	434.39
	June	Peak Season	90.86	488.75	76711099	447.03
	July	Peak Season	89.35	474.63	75832051	427.66
	December	Peak Season	88.92	471.98	75143471	423.77
	April	Shoulder Season	73.3	355.77	45382422	264.47
	March	Shoulder Season	71.58	346.50	44636545	251.73
	May	Shoulder Season	70.95	342.52	43698481	246.44
	November	Shoulder Season	70.66	344.45	42372474	246.93
	October	Low Season	53.2	275.66	26543659	149.69
	January	Low Season	53.06	275.61	26422176	149.01
	September	Low Season	52.85	278.25	25739081	149.99
	February	Low Season	52.61	274.30	24034752	147.43

Analysis 3: During which periods should Hilton increase or decrease room prices?

Objective: Identify pricing opportunities based on occupancy and demand.

Insight: Very High demand periods achieved **94.12% occupancy** and a **\$529.09 ADR**, supporting price increases, whereas Low demand periods recorded only **45.02% occupancy** and **\$249.83 ADR**, indicating the need for promotions.

	Season	Demand_Level	Average_Occupancy_Percentage	Average_ADR	Average_RevPAR	Number_of_Days	Pricing_Strategy
►	Peak Season	Very High	94.12	529.09	497.9	152	Increase ADR Opportunity
	Shoulder Season	High	82.75	404.28	334.4	89	Increase ADR Opportunity
	Peak Season	High	82.57	397.47	328.32	94	Increase ADR Opportunity
	Shoulder Season	Medium	65.23	314.52	205.21	155	Maintain Pricing Strategy
	Low Season	Medium	64.7	314.77	203.37	97	Maintain Pricing Strategy
	Low Season	Low	45.02	249.83	112.48	144	Discount / Promotion Opportunity

Analysis 4: What is the relationship between occupancy rate and ADR?

Objective: Examine the relationship between occupancy and pricing.

Insight: ADR increased from **\$249.54** at **42.8% occupancy** to **\$459.00** at **87.85% occupancy**, while RevPAR nearly quadrupled (**\$106.82 → \$406.88**), confirming a strong positive relationship between occupancy and pricing.

	Occupancy_Category	Average_Occupancy_Percentage	Average_ADR	Average_RevPAR	Number_of_Days
►	High Occupancy	87.85	459.00	406.88	335
	Low Occupancy	42.8	249.54	106.82	112
	Medium Occupancy	63.64	307.43	196.36	284

Analysis 5: How does Hilton's ADR compare against competitors?

Objective: Compare Hilton's pricing position with competing hotels.

Insight: Hilton's average ADR (**\$368.02**) remained within **±\$1** of five leading competitors, indicating a strong and competitive market pricing strategy.

	Competitor_Hotel	Average_Competitor_ADR	Average_Hilton_ADR	Average_Price_Difference	Hilton_Position
►	Marriott Waikiki Beach	368.17	368.02	-0.15	Hilton Priced Lower
	Hyatt Regency Waikiki	368.15	368.02	-0.13	Hilton Priced Lower
	Outrigger Waikiki Beach	367.85	368.02	0.18	Hilton Priced Higher
	Sheraton Waikiki	367.48	368.02	0.55	Hilton Priced Higher
	Alohilani Resort	367.11	368.02	0.91	Hilton Priced Higher

Analysis 6: Which room categories have the biggest pricing opportunities compared to competitors?

Objective: Identify room categories requiring pricing adjustments.

Insight: Ocean View, Standard, and Ocean Front Rooms all remained within **\$0.61** of competitor ADRs, suggesting Hilton's room pricing is well aligned with the market while leaving limited room for adjustment.

	Room_Type	Average_Competitor_ADR	Average_Hilton_ADR	Pricing_Gap	Pricing_Action
►	Ocean View Room	368.15	368.02	0.13	Market Competitive
	Standard Room	367.69	368.02	-0.34	Market Competitive
	Ocean Front Room	367.41	368.02	-0.61	Market Competitive

Analysis 7: Do major events increase hotel demand?

Objective: Evaluate the impact of market events on occupancy and ADR.

Insight: Major events drove occupancy as high as **90.68%** and ADR up to **\$492.00**, with Cultural Festivals, Surf Competitions, and Independence Day generating the strongest demand.

	Event_Name	Event_Type	Expected_Demand	Average_Occupancy_Percentage	Average_ADR	Average_RevPAR	Event_Days
►	Independence Day Celebration	Holiday	Very High	89.95	492.00	444.29	2
	Christmas Holiday	Holiday	Very High	80.93	397.50	322.59	2
	New Year Celebration	Holiday	Very High	62.27	305.50	189.68	2
	Cultural Festival	Cultural Event	Medium	90.68	478.00	439.73	2
	Hawaii Food Festival	Cultural Event	Medium	85.31	477.00	409.06	2
	Surf Competition	Conference	Medium	83.66	434.00	377.38	3
	Tourism Expo	Cultural Event	Medium	82.28	365.00	303.53	2
	Tourism Expo	Conference	Medium	82.26	441.00	383.86	2
	Cultural Festival	Festival	Medium	80.24	409.00	334.22	3
	Tourism Expo	Festival	Medium	77.79	396.00	321.84	3
	Medical Conference	Festival	Medium	74.76	410.33	324.55	3
	Medical Conference	Conference	Medium	72.99	364.20	275.6	5
	Cultural Festival	Sports	Medium	71.7	378.33	299.42	3
	Music Festival	Cultural Event	Medium	70.88	342.50	256.4	4
	Business Conference	Cultural Event	Medium	67.05	418.00	323.62	2

Analysis 8: Which events should Hilton prioritize for revenue optimization?

Objective: Identify events generating the strongest revenue opportunities.

Insight: Cultural Festivals generated the highest revenue (**\$16.48M**), followed by Tourism Expos (**\$16.05M**) and Surf Competitions (**\$15.48M**), making them the most valuable recurring revenue opportunities.

	Event_Name	Total_Revenue	Average_Occupancy_Percentage	Average_ADR	Average_RevPAR	Event_Days	Revenue_Category
►	Business Conference	9209232	75.34	397.10	322	10	Moderate Revenue Opportunity
	Christmas Holiday	1845195	80.93	397.50	322.59	2	Moderate Revenue Opportunity
	Cultural Festival	16483531	78.06	410.76	339.03	17	Moderate Revenue Opportunity
	Hawaii Food Festival	13504486	76.69	413.71	337.27	14	Moderate Revenue Opportunity
	Independence Day Celebration	2541342	89.95	492.00	444.29	2	High Revenue Opportunity
	Medical Conference	13281703	69.77	352.83	258	18	Low Revenue Opportunity
	Music Festival	8116473	62.01	308.93	202.71	14	Low Revenue Opportunity
	New Year Celebration	1084987	62.27	305.50	189.68	2	Low Revenue Opportunity
	Spring Break	1140730	65.6	304.50	199.43	2	Low Revenue Opportunity
	Surf Competition	15477611	75.25	377.61	300.65	18	Moderate Revenue Opportunity
	Thanksgiving Holiday	1917678	83.48	403.50	335.26	2	Moderate Revenue Opportunity
	Tourism Expo	16051975	74.21	373.47	295.4	19	Moderate Revenue Opportunity

Analysis 9: What guest experience factors impact satisfaction the most?

Objective: Determine which service areas most influence guest ratings.

Insight: Check-in Experience and Cleanliness achieved the highest average satisfaction scores (**3.72/5**), while Food & Dining (**3.64/5**) and Location (**3.66/5**) presented the greatest improvement opportunities.

	Review_Category	Total_Reviews	Average_Rating	Positive_Reviews	Negative_Reviews	Positive_Sentiment_Percentage
►	Check-in Experience	1299	3.72	847	193	65.20
	Cleanliness	1200	3.72	788	171	65.67
	Room Quality	1299	3.71	852	190	65.59
	Staff Service	1228	3.70	818	194	66.61
	Value for Money	1186	3.70	769	169	64.84
	Amenities	1282	3.67	836	203	65.21
	Location	1307	3.66	823	212	62.97
	Food & Dining	1199	3.64	761	188	63.47

Analysis 10: Does guest satisfaction influence booking value?

Objective: Analyze the relationship between guest sentiment and booking value.

Insight: Positive guest reviews generated **\$10.37M** in booking revenue compared to **\$2.35M** from negative reviews, with average booking values of **\$1,596.71** versus **\$1,542.76**, demonstrating the financial impact of guest satisfaction.

	Sentiment	Average_Rating	Total_Reviews	Average_Booking_Revenue	Average_Length_of_Stay	Total_Revenue_Generated
►	Positive	4.38	6494	1596.71	4.51	10369039
	Neutral	3.00	1986	1581.33	4.42	3140526
	Negative	1.66	1520	1542.76	4.42	2345000

5. Power BI Dashboard

An interactive Revenue Management Dashboard was developed in Power BI to provide a centralized view of hotel performance and support data-driven pricing decisions.

Dashboard KPIs

- Total Revenue
- Total Bookings
- Average Length of Stay
- Average Occupancy Rate

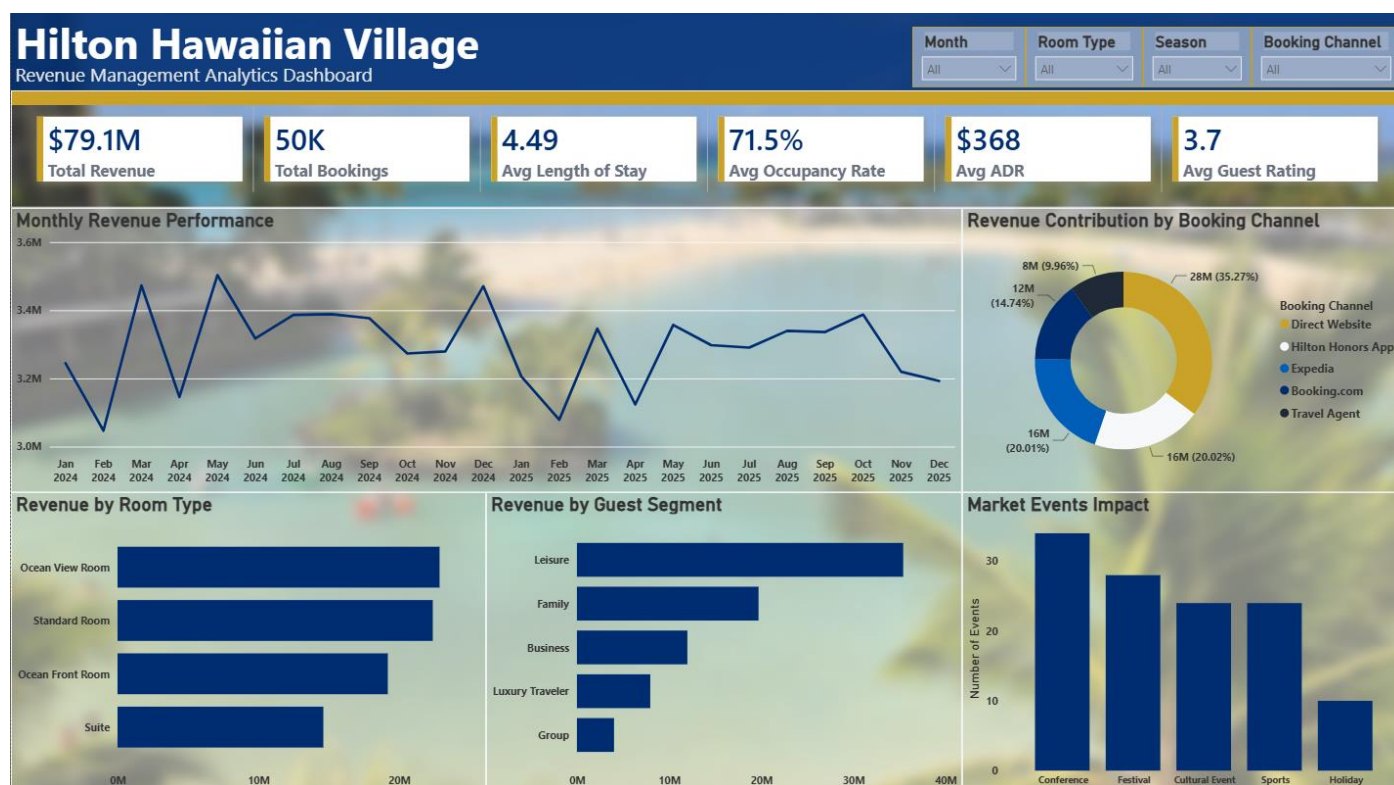
- Average ADR
- Average Guest Rating

Interactive Filters

- Booking Month
- Room Type
- Season
- Booking Channel

Dashboard Visualizations

- Monthly Revenue Performance
- Revenue Contribution by Booking Channel
- Revenue by Room Type
- Revenue by Guest Segment
- Market Events Impact



6. Key Findings

- Ocean View Rooms generated the highest revenue (\$22.87M, 28.92% of total revenue), while Suites achieved the highest ADR (\$734.11), indicating strong premium pricing potential despite lower booking volume.
- Peak Season (June-August and December) recorded occupancy above 88%, ADR between \$472-\$489, and monthly revenue exceeding \$75M, whereas Low Season occupancy dropped to nearly 53% with revenue below \$27M.

- Very High demand periods achieved 94.12% occupancy and a \$529.09 ADR, while Low demand periods fell to 45.02% occupancy and \$249.83 ADR, validating dynamic pricing opportunities.
 - High occupancy periods generated an average ADR of \$459 compared to only \$249 during low occupancy periods, demonstrating a strong positive relationship between occupancy and pricing power.
 - Hilton's average ADR (\$368.02) remained within $\pm\$1$ of five major competitors, reflecting a well-balanced and competitive pricing strategy.
 - Major events significantly increased hotel performance, with Cultural Festivals generating \$16.48M in revenue and Independence Day achieving 89.95% occupancy with a \$492 ADR.
 - Guest satisfaction directly influenced revenue, with positive reviews producing \$10.37M in booking value compared to \$2.35M from negative experiences, while Check-in Experience received the highest satisfaction score (3.72/5).
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7. Business Recommendations

1. Implement Demand-Based Dynamic Pricing

Increase ADR during peak seasons, holidays, and high-demand events where occupancy consistently exceeds 88%, while introducing targeted discounts and promotional packages during low-demand periods to improve occupancy without significantly reducing profitability.

2. Maximize Revenue from Premium Room Categories

Focus marketing campaigns, upgrade offers, and bundled packages on Ocean View Rooms and Suites. Ocean View Rooms contributed nearly 29% of total revenue, while Suites generated the highest ADR, making them the strongest revenue drivers.

3. Develop Event-Focused Revenue Strategies

Create dedicated pricing, marketing, and inventory plans around Cultural Festivals, Tourism Expos, Surf Competitions, and national holidays. These events consistently produced the highest occupancy and revenue, presenting opportunities to maximize RevPAR through early demand forecasting.

4. Continuously Benchmark Competitor Pricing

Although Hilton's ADR closely matches competitors, regular monitoring of competitor pricing should be integrated into revenue management decisions to capture additional pricing opportunities while maintaining market competitiveness.

5. Prioritize Improvements in Guest Experience

Invest in operational improvements for check-in efficiency, room quality, cleanliness, and staff service. Positive guest experiences generated more than four times the booking revenue of negative experiences, demonstrating that service quality directly contributes to higher customer value and repeat business.

8. Conclusion

This project demonstrates how data analytics can support strategic revenue management within the hospitality industry. Using Python for data preparation, MySQL for business analysis, and Power BI for interactive visualization, the project transformed hotel operational data into meaningful business insights. The findings provide practical recommendations for pricing optimization, demand forecasting, customer experience enhancement, and revenue growth, illustrating the value of data-driven decision-making in hotel management.